DEEPAK SHARMA

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TECHNICAL SKILLS

Languages: Python, C/C++, SQL, PHP, HTML/XML
Frameworks & Libraries: Flask, NumPy, Pandas, Matplotlib, OpenCV, C++ STL
Databases & Tools: MySQL, MongoDB, Git, Linux, Raspberry Pi
Cybersecurity & ICS: OpenPLC, Modbus TCP, Conpot honeypot, S7Comm, BACnet, SPAN port mirroring, TCP SYN/Flooding attacks
Core CS & Analytics: Data Structures & Algorithms, Operating Systems, OOP, DBMS, Computer Networks, NLP, Excel

WORK EXPERIENCE

Research Intern – Indian Institute of Science (IISc) *Jan 2026 – May 2026*

- Deployed an ICS/OT security testbed using OpenPLC, Raspberry Pi, and Modbus TCP with SPAN port mirroring.
- Executed Modbus flooding, TCP SYN, and payload injection attacks to quantify OT service disruption.
- Built a custom Python ICS attack suite integrated with multi-IED Conpot honeypots (Modbus, S7Comm, BACnet) for passive threat intelligence capture.

Software Development Engineer Intern – Adysoft *Dec 2024 – Feb 2025*

- Built and optimized a PHP-based online billing application with scalable, reusable modules.
- Improved application performance and reduced billing cycle errors.
- Refactored legacy code with the development team, improving maintainability and user experience.

Cybersecurity Intern – CDAC (Centre for Development of Advanced Computing) *Jun 2024 – Jul 2024*

- Designed and applied security measures to strengthen data protection and system reliability.
- Partnered with senior engineers to analyze and mitigate active threat vectors, contributing to a 40% reduction in identified vulnerabilities.

PROJECTS

MediCrypt Lattice — Medical Image Encryption — GitHub

Tech: Python, Flask, PWLCM, Lattice Cryptography, NumPy, SHA-256

• Engineered a secure medical image encryption system using hybrid PWLCM–Lattice cryptography with chaotic permutation and lattice-based keystream generation.

• Built a Flask-based end-to-end pipeline for image upload, encryption, decryption, hash validation, and histogram analysis, achieving lossless recovery.

TwitPersona — AI Tweet Reply Chrome Extension — GitHub

Tech: JavaScript, Chrome Extension APIs, Google Gemini API

• Built a Chrome extension generating persona-driven, AI-powered tweet replies using the Google Gemini API for automated, context-aware responses.

• Executed real-time DOM manipulation and Chrome storage APIs for persistent persona configuration and cross-session settings.

EncryptEase — Secure File Encryption Tool — GitHub

Tech: C++, OpenSSL, File Security

• Designed a secure file encryption and storage tool using OpenSSL with SHA-256 password hashing and Base64 encoding for authentication.

• Established user authentication with masked password input, password recovery, and role-based access control with secure credential storage across Windows, Linux, and macOS.

EDUCATION

B.Tech, CSE — AI & Data Science | Indian Institute of Information Technology, Senapati Manipur (IIITM) *Nov 2022 – May 2026*
Cumulative CGPA: 8.66 / 10

All India Senior School Certificate (Class XII) | Saraswati Vidya Mandir Sr. Secondary School - Meham, Rohtak *Apr 2019 – Mar 2020*
Score: 94.2%

ACHIEVEMENTS & LEADERSHIP

- Academic Excellence & Branch Transition: Ranked 1st in department in Year 1, earning a merit-based branch change from ECE (VLSI) to CSE (AI & DS).
- Lead, Qurbat Poetry Club — IIIT Senapati: Mentored 50+ students and organized poetry workshops and campus events.
- Public Relations Co-Lead, Techfest Ahouba 2.0 — IIIT Senapati: Led digital outreach and artist onboarding, driving 5,000+ event engagements.